

anywhere and made accessible to everyone in the system. Blockchain can help remedy this situation.

Blockchain technology can help track a drug from the time of its inception as an idea. So right from the nascent stages of animal testing, results of tests can be recorded on the blockchain and be available to the doctor who is prescribing the medicine to his/ her patient. Since the information is time stamped and tamper-free, the source of origin is known to everyone in the chain. This becomes even more important in the case of customised medication. While the idea of customising medicines based on gene therapy is not as widespread today, it could well be the face of modern medicine in the years to come.

For instance, white blood cells from Patient A can be sent from the hospital to a lab miles away to create a medication suited specifically to A's genetic type. The creation of the drug would depend on the white blood cells being transported and shuffled back and forth between a number of labs. If there is no clear traceability, the drug that becomes available could be ineffective for Patient A, could be a counterfeit, or worse could be based on white blood cells taken from Patient B (patient sample mix ups). Blockchain could help put the drug's entire lifecycle in blocks that are viewable by everyone involved in the procedure of creating the drug.

Another extremely important aspect of drug traceability is stopping counterfeit drugs from entering the supply chain. Given that everything in blockchain is time-stamped and immutable, tampering is impossible. This would mean that fraudsters couldn't unload real drugs and replace them with counterfeit drugs without the system catching it. Even if they did try, the crime could be traced back to them. Also, in case of delayed shipments, blockchain could help in determining which party last handled the drugs. This will help in speeding up the process of the drugs reaching their intended destination.

Clinical trial

The process of getting a new drug into the market is usually an extremely long and tedious one. Before a drug receives approval from the regulatory board, it has to undergo various phases of testing. Right from testing on animals to finally being tested on humans, the collection of data is a logistical nightmare for everyone involved. Let's not even begin to debate the chances of data tampering that can happen when there are multiple levels of personnel involved in research that is worth millions or more. While